

## Gradient function graphs



1

- a 1
- b 1
- c A linear function has a constant gradient.

2

- a The gradient function is positive when  $y = x^2$  is increasing, and negative when  $y = x^2$  is decreasing.
- b Positive
- c The gradient of the tangent line increases at a constant rate.
- d Negative
- e The gradient of the tangent line increases at a constant rate.
- f Linear

3

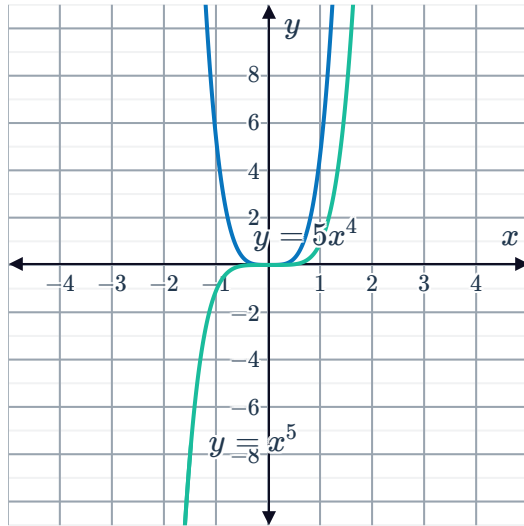
- a Positive
- b The gradient of the tangent line increases at an increasing rate.
- c Positive
- d The gradient of the tangent line decreases at an decreasing rate.
- e Quadratic

4

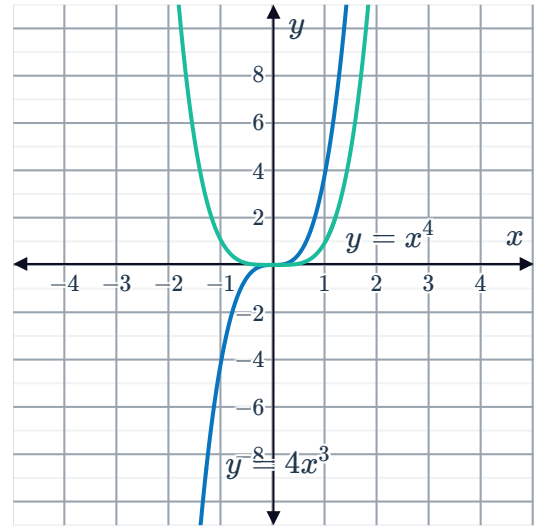
- a Positive
- b The gradient of the tangent line increases at an increasing rate.
- c Negative
- d The gradient of the tangent line increases at a decreasing rate.
- e Cubic

5

a



b



c Odd

d Even

## Algebraic differentiation

6

a

i Degree 1

ii Degree 2

iii Degree 3

b  $n - 1$ 

7

a

$$\frac{dy}{dx} = 7x^6$$

b

$$\frac{dy}{dx} = 9x^8$$

c

$$\frac{dy}{dx} = 5x^4$$

d

$$\frac{dy}{dx} = 0$$

## Applications

8

$$f'(2) = 32$$

9

a 9

$$b \quad y = 9x - 8$$

10



a

| Graph     | $y$ -intercept of the tangent line | Gradient of the tangent |
|-----------|------------------------------------|-------------------------|
| $y = x^2$ | $(0, -1)$                          | 2                       |
| $y = x^3$ | $(0, -2)$                          | 3                       |
| $y = x^4$ | $(0, -3)$                          | 4                       |
| $y = x^5$ | $(0, -4)$                          | 5                       |

b  $n$ c No, because then the gradient of the tangent at  $x = 1$  would be 1.

11

a False

b True

c False

d True